PARKING SLOT ALLOCATION USING RFID TECHNOLOGY

20RM600 RESEARCH METHODOLOGY

BY

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Introduction

- One of the most popularly encountered problems in recent days is parking the vehicles
- The parking lot area has many lanes/slots. Moreover, this involves a lot of manual labor.
- Conventional parking systems do not have any intelligent monitoring systems.
- Here, an efficient parking management system is proposed using RFID Technology to bring down the waiting time.

Radio Frequency Identification (RFID)

Cost effective

Passive RFID card – High durability

Complex encryption algorithms

Dustproof & Waterproof

Literature Survey

RFID & GSM Feature

- ❑ Pre-booking the slot by application with OTP **[2][5][7].**
- ❑ Usage of RFID Tags and RFID Reader



RFID Passive TAG



RFID Active TAG



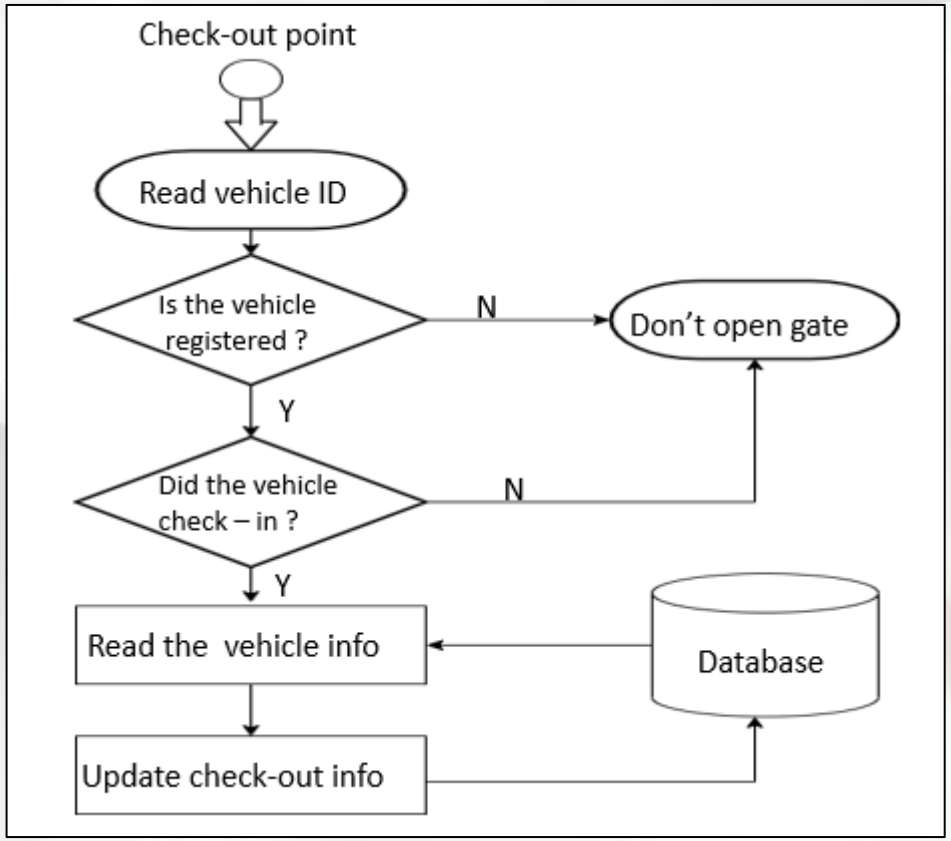
RFID Reader Module

Source: Google

Literature Survey

Security

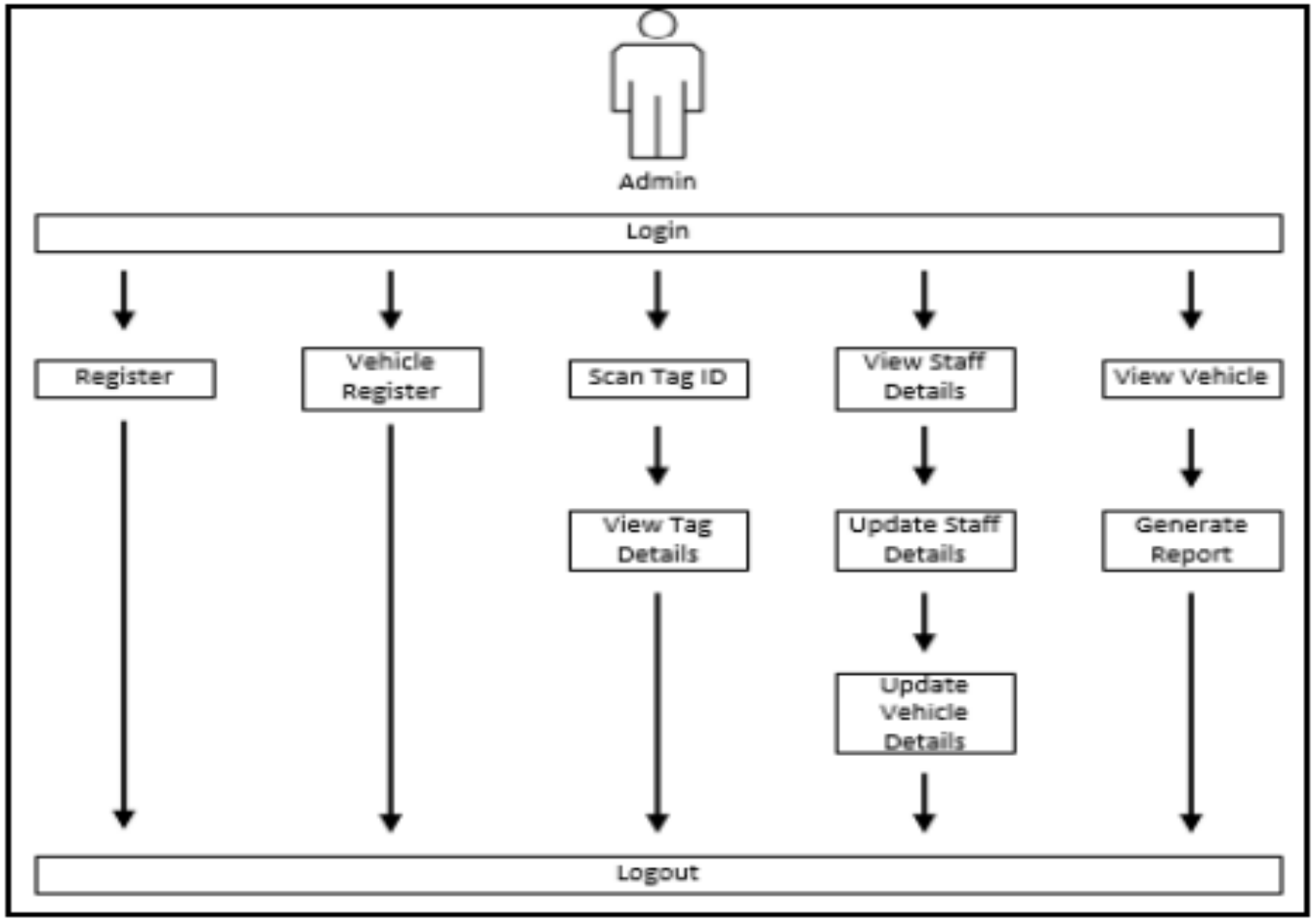
- Check-ins and check-outs are controlled by the software on the basis of smart card **[4][8]**.



Parking lot check-out process [1]

Literature Survey

Registration Process in User Interface



Block diagram of Registration Process [4]

Literature Survey

Parking Charges

- Deducts the charges at the exit **[6].**
- The real-time detection of improper parking and automatic collection of parking charges are developed using E-parking system **[3][10].**



Exit Gate of the Parking Lot

Source: Google

Research Gap

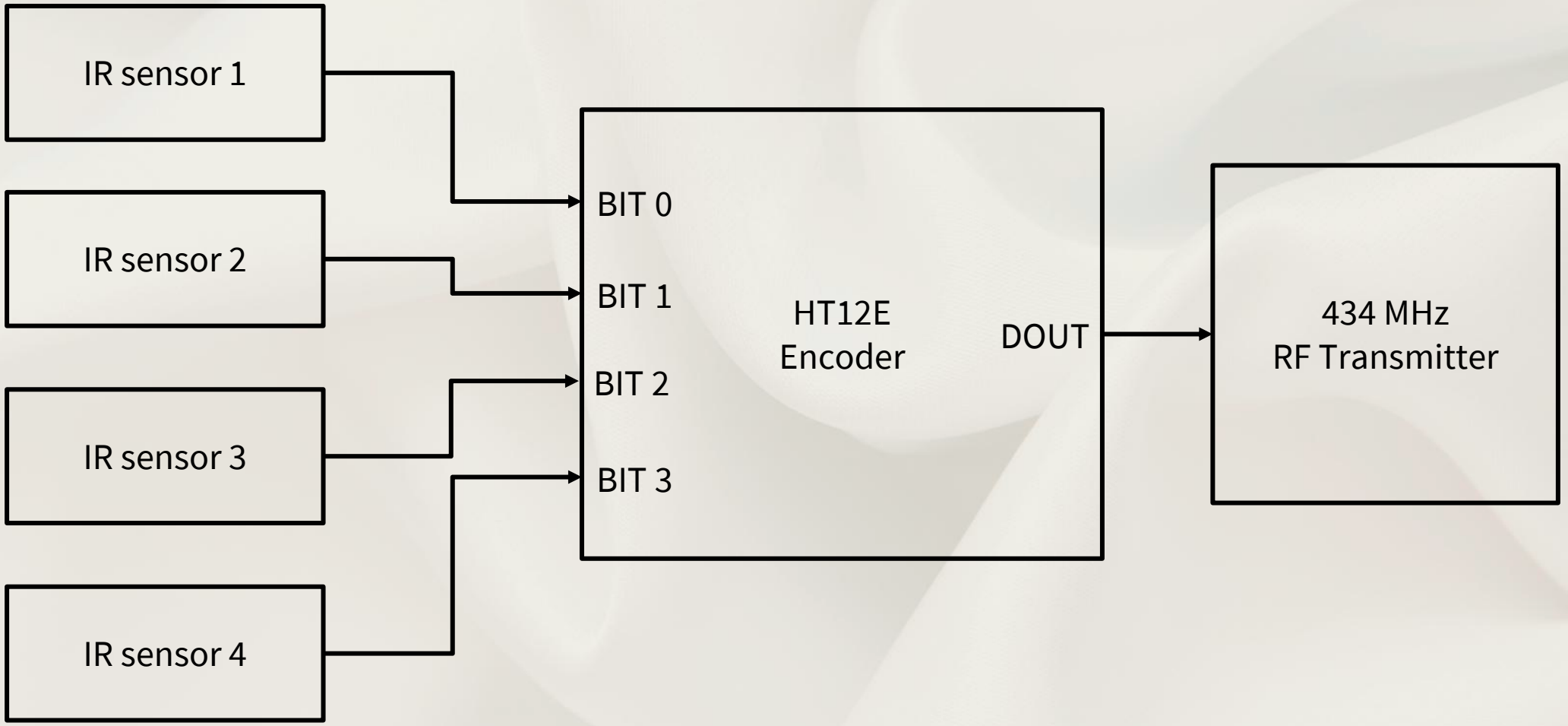
- Notifying the user about the details of parked vehicles
- The Management Costs can be decreased by automating the parking management system
- Payments and POS can be integrated
- The parking solution can be optimized by managing the traffic and there by reducing pollution

Objectives

To build a smart parking prototype using RFID Technology

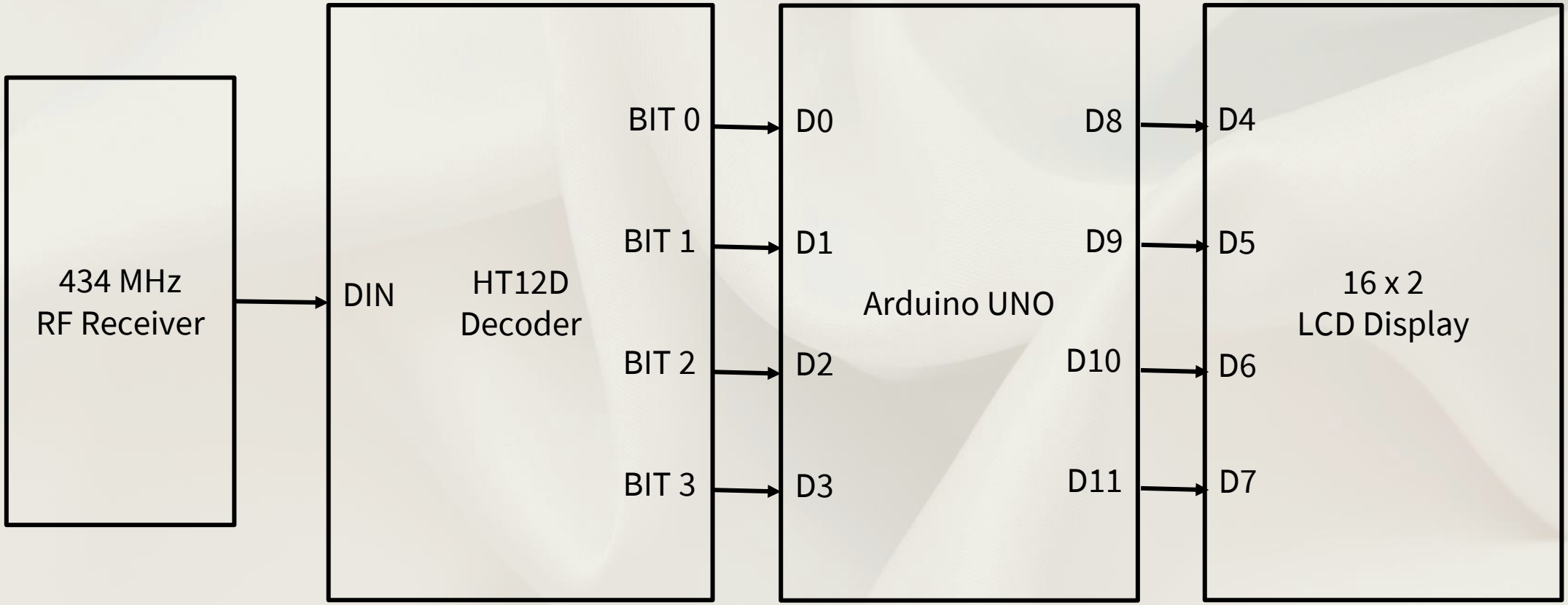
- ❑To find the empty slots in parking lot
- ❑To notify the user about the allocated slot via SMS
- ❑To deduct the charges of parked vehicles at the exit

Methodology



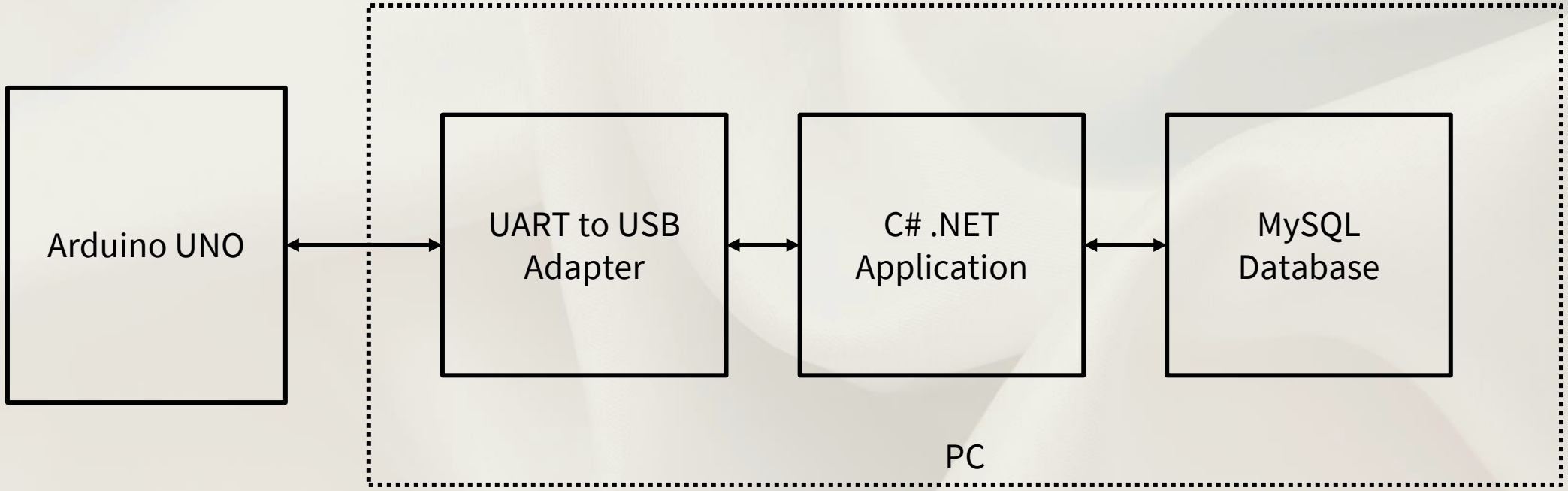
Pin diagram of Transmitter Circuit

Methodology (Cont'd)



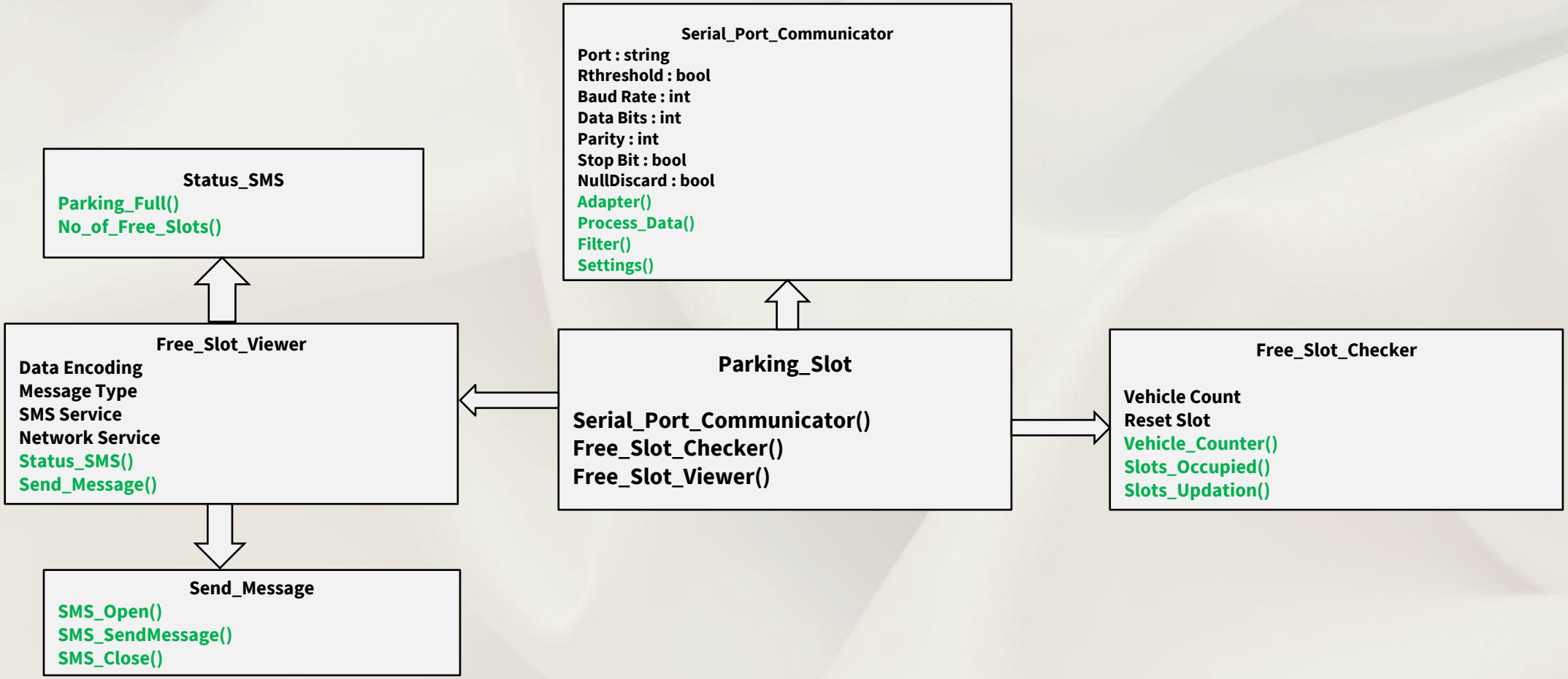
Pin diagram of Receiver Circuit

Methodology (Cont'd)



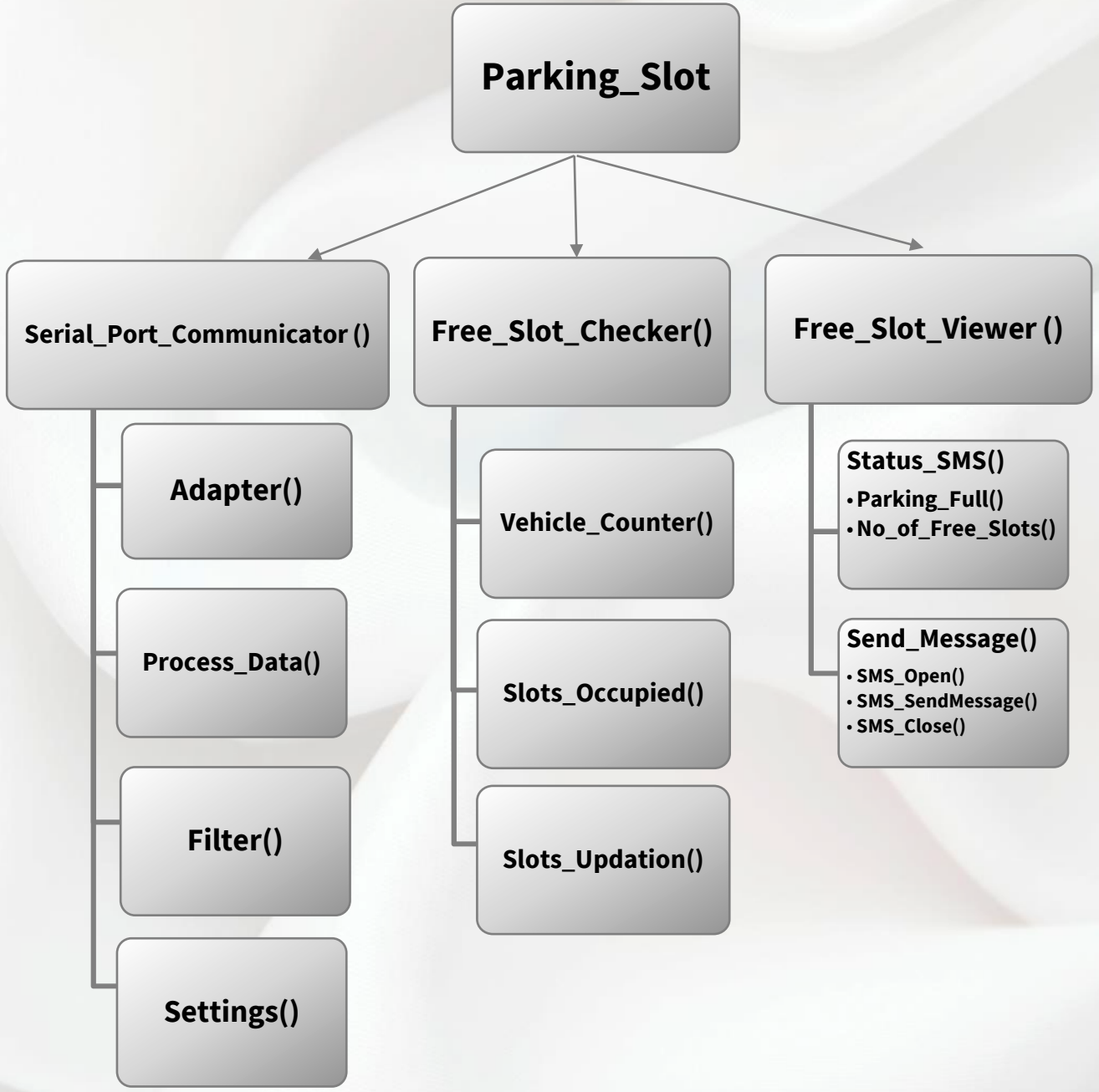
Database Interface Block Diagram

Software Implementation



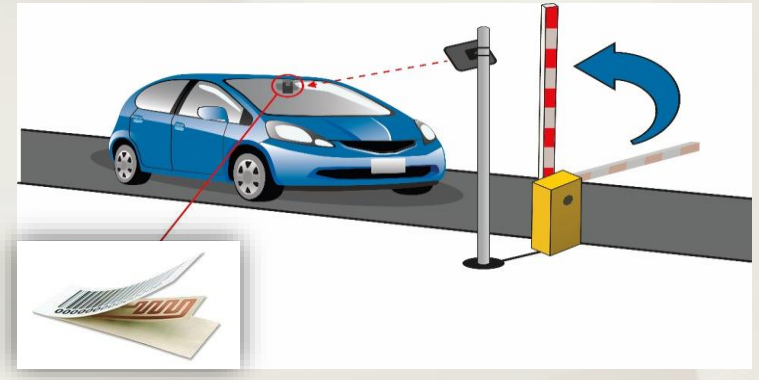
Class diagram for parking slot allocation system

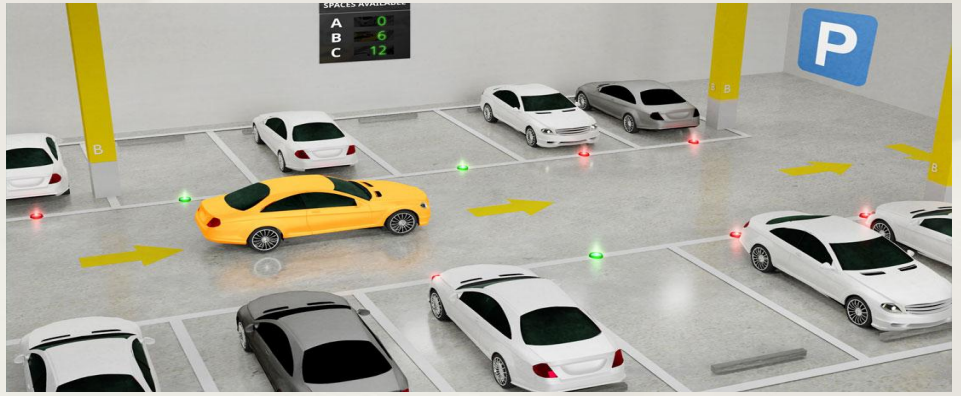
Software Implementation (Cont'd)



Workflow of Class diagram

Overview of Smart Parking System







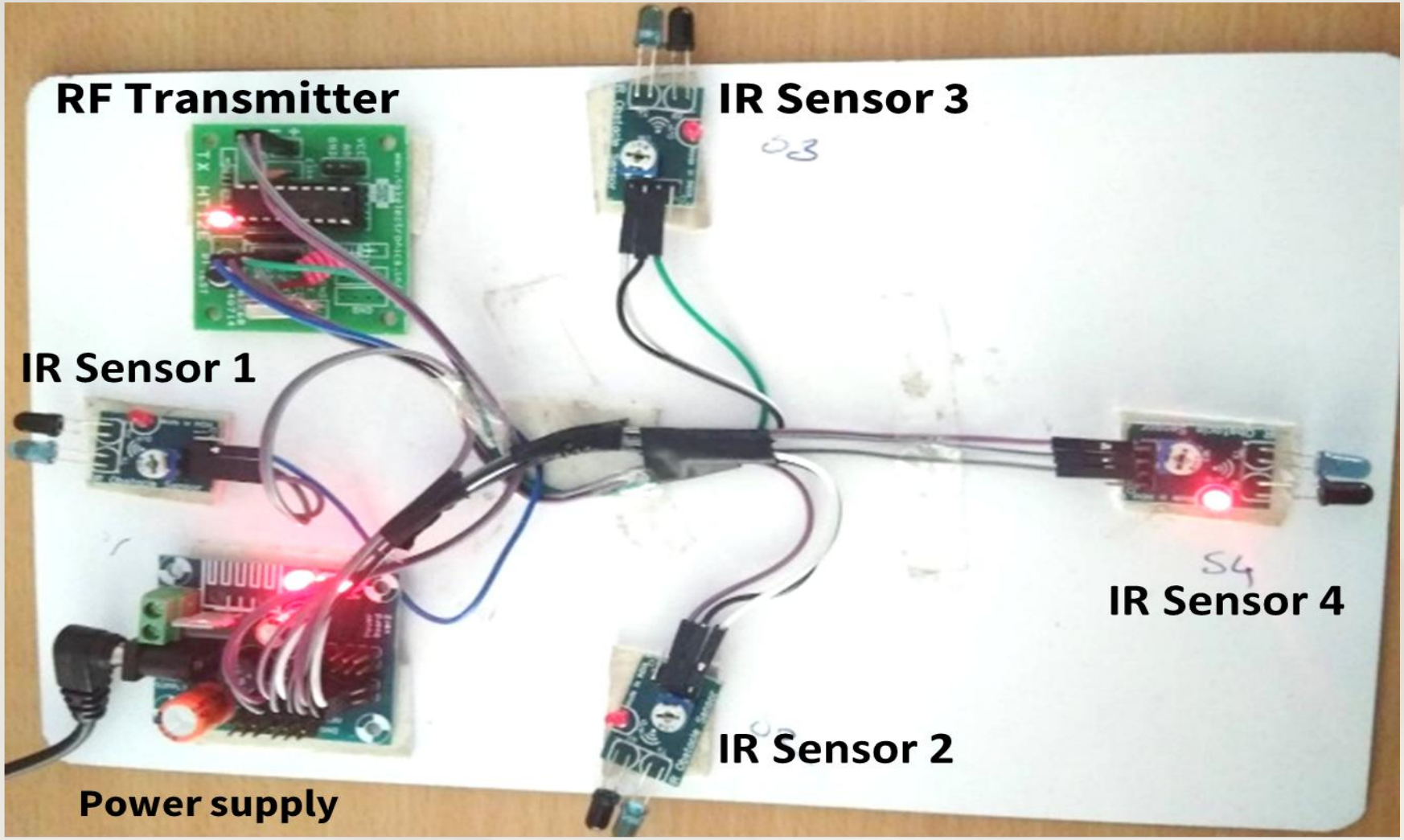




Overview of the system

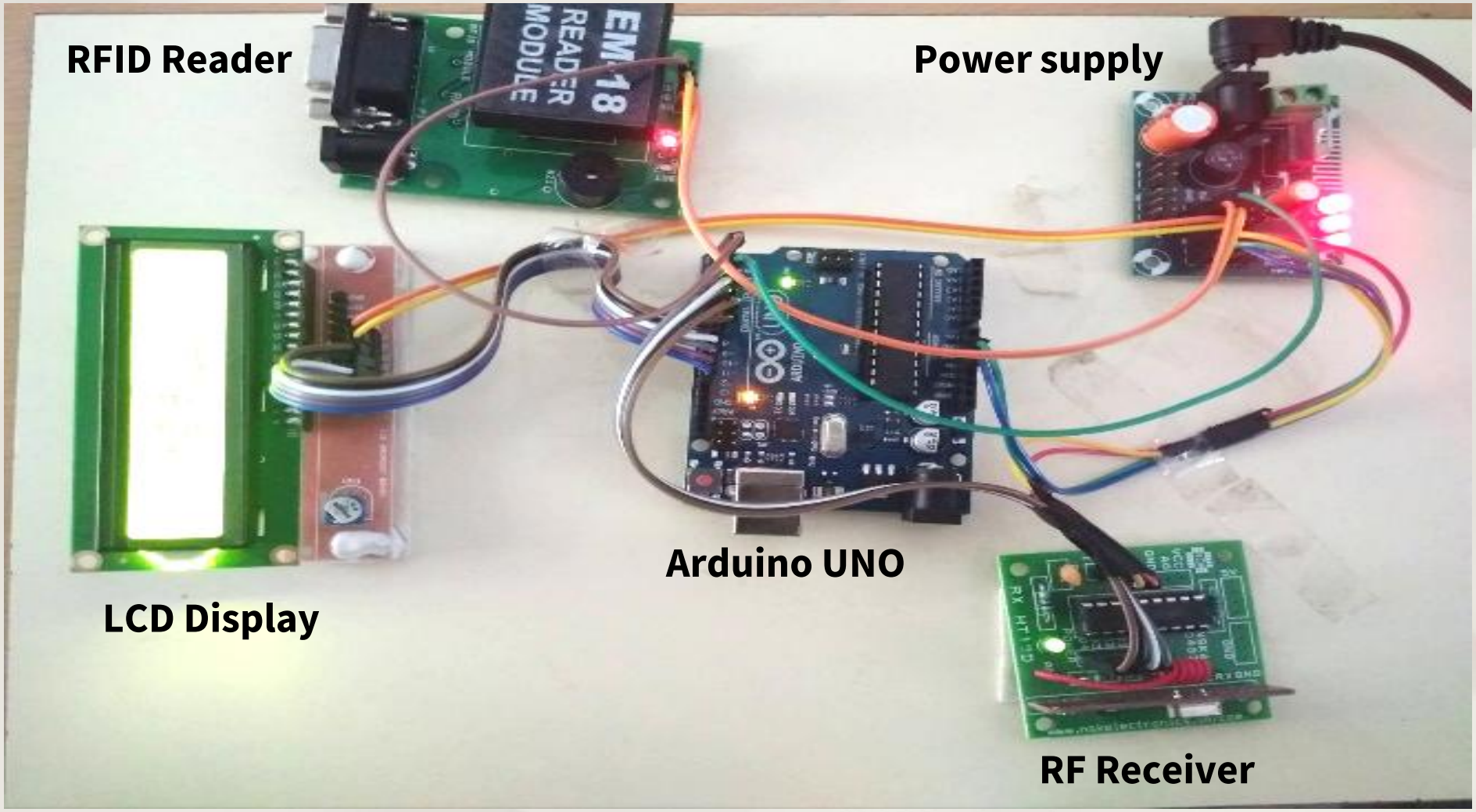
Source: Google

Results



Parked status transmission System

Results (Cont'd)



Conclusion

- The prototype of RFID-based Smart parking system is successfully built
- The prototype mimics the small smart parking scenario
- Finding and allocating the empty slots in parking lot
- User is notified about the parked vehicle details via SMS

Future Work

- This work can be further carried out by calculating and deducting the charges of the parked vehicles at the exit of the parking lot
- Connect all parking areas in a city using a single web server
- Promote the use of autonomous, smart and non-polluting vehicles to reduce traffic and pollution.

References

[1] Z. Pala and N. Inanc, “Smart Parking Applications Using RFID Technology,” 2007 1st Annual RFID Eurasia, 2007, pp. 1-3, doi: 10.1109/RFIDEURASIA.2007.4368108

[2] K. Ganesan and K. Vignesh, "Automated parking slot allocation using RFID technology," 2007 9th International Symposium on Signal Processing and Its Applications, 2007, pp. 1-4, doi: 10.1109/ISSPA.2007.4555296

[3] H.Tanti , Pratik Kasodariya , Shikha Patel , Dhaval H Rangrej, 2020, Smart Parking System based on IOT, International Journal Of Engineering Research & Technology (IJERT) Volume 09, Issue 05 (May 2020).

[4] Mazlan, Muhammad & A Hamid, isredza rahmi & Kamaludin, Hazalila. (2018). Radio Frequency Identification (RFID) Based Car Parking System. JOIV : International Journal on Informatics Visualization. 2. 10.30630/joiv.2.4-2.173.

[5] Zala, Yuvraj & Patel, Akash & Makwana, Deval & Botta, Ravi. (2017). Implementation of cost-effective RFID based smart parking management system with enabled GSM feature.(ICCMIT 2017) Warsaw, Poland.

References

[6] Sadhukhan, Pampa. (2017). An IoT-based E-Parking System for Smart Cities. 10.1109/ICACCI.2017.8125982.

[7] Sahil Kalpesh Desai “Smart Car Parking System” International Advanced Research Journal in Science, Engineering and Technology .Vol. 7, Issue 1, January 2020.

[8] L. Kumar, M. H. Khan and M. S. Umar, "Smart parking system using RFID and GSM technology," 2017 International Conference on Multimedia, Signal Processing and Communication Technologies (IMPACT), 2017, pp. 180-184, doi: 10.1109/MSPCT.2017.8364000.

[9] E. h. ROUAN, S. SAFI and A. BOUMEZZOUGH, "An automated parking access control system based on RFID technology," 2020 15th Design & Technology of Integrated Systems in Nanoscale Era (DTIS), 2020, pp. 1-2, doi: 10.1109/DTIS48698.2020.9080913.

[10] Mudaliar, Shruthi & Agali, Shreya & Mudhol, Sujay & Jambotkar, Chaitanya. IoT Based Smart Car Parking System. IJSART - Volume 5 Issue 1 – Jan 2019

THANK YOU